
TENSOR NETWORK REPRESENTATION OF FOURIER TRANSFORMS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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We derive two standard algorithms in the contraction formalism:

- Fast Fourier Transform
- Quantum Fourier Transform

1 Definition

The Fourier transform of vectors in $\mathbb{C}^m$ is represented by a matrix

$$F[X = x, Y = y] = \frac{1}{\sqrt{m}} \exp \left[2\pi i \cdot \frac{x \cdot y}{m} \right]$$

where X, Y are variables of dimension m . The Fourier transform of a vector $V[X]$ is then the contraction

$$\langle V[X], F[X, Y] \rangle_{[Y]} .$$

1.1 Unfolding

We assume $m = 2^d$ and represent each index $x \in [2^d]$ by a tuple $(x_0, \dots, x_{d-1}) \in [2]^d$ such that

$$x = \sum_{k \in [d]} x_k \cdot 2^{d-k-1} .$$

Note that in this convention, x_{d-1} is the least significant bit. The unfolded vector $V[X]$ is a tensor $V[X_{[d]}]$ with indices

$$V[X_{[d]} = x_{[d]}] = V[X = \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1} \right)] .$$

In the unfolded representation the fourier transform is described as the contraction with a tensor $F^d[X_{[d]}, Y_{[d]}]$ with coordinates

$$F^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \frac{1}{\sqrt{2^d}} \exp \left[2\pi i \cdot \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1} \right) \cdot \left(\sum_{k \in [d]} y_k \cdot 2^{d-k-1} \right) \right] ,$$

that is the Fourier transform of $V[X_{[d]}]$ is

$$\langle V[X_{[d]}], F[X_{[d]}, Y_{[d]}] \rangle_{[Y_{[d]}]} .$$

2 Tensor network decomposition

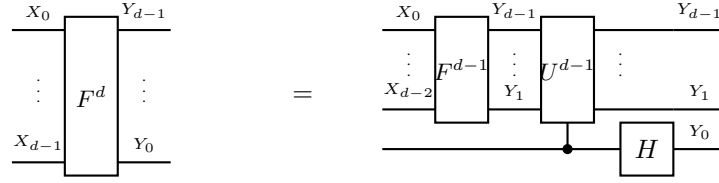
We now derive a tensor network decomposition of the tensor $F^d[X_{[d]}, Y_{[d]}]$.

To this end we introduce the diagonal unitary

$$U^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \begin{cases} \exp \left[2\pi i \cdot \left(\sum_{k \in [d]} x_k \cdot 2^{-k-2} \right) \right] & \text{if } x_{[d]} = y_{[d]} \\ 0 & \text{else} \end{cases}$$

Note that U^d is diagonal and the tensor product of matrices with variables X_k, Y_k . Diagonality can be used to contract U^d to an arbitrary tensor with computational demand in $\iota(2^d)$.

With these, the Fourier transform on $[2^d]$ can be decomposed into a Fourier transform on $[2^{(d-1)}]$ as:



By iteratively decomposing F^{d-1} until we arrive at F^1 (which is H), we get a decomposition of F^d into d H matrices and controlled U unitaries.

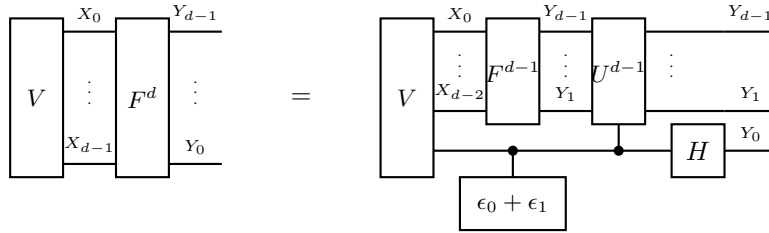
3 Fast Fourier Transform

3.1 Layerwise contraction

We contract the FFT tensor network layerwise. There are $2 \cdot d$ layers, each has a contraction demand in $\mathcal{O}(2^d)$ (exploiting diagonality of U). The algorithm has a total computational demand in $\mathcal{O}(d2^d)$.

3.2 Divide and conquer

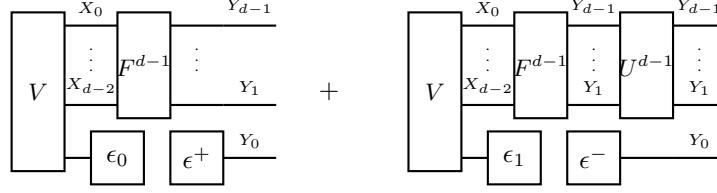
Called the radix-2 FFT Cooley and Tukey (1965). Can be seen as an iterative simplification by adding a $\mathbb{I}[X_{d-1}]$ and splitting it into the basis vectors.



Now by linearity of contractions, this splits into two terms



We simplify them to



The contraction of V with $\epsilon_1 [X_{d-1}]$ respectively $\epsilon_1 [X_{d-1}]$ are the even and the odd coordinates of V . Both are Fourier transformed with $d - 1$ bits, which are further divided into Fourier transforms of lower order. In the conquer step of the FFT, the odd Fourier transform gets a coordinatewise transform by U , which is efficient since U is diagonal.

4 Quantum Fourier Transform

When $V[X_{[d]}]$ is a quantum state, the Fourier transform can be understood as a unitary transforming it. Since both the controlled U tensor and the Hadamard H are unitary, each of them can be implemented by a sequence of gates. For sparse implementation one can exploit that U is a tensor product of matrixes (i.e. single qubit gates).

References

James W. Cooley and John W. Tukey. An algorithm for the machine calculation of complex Fourier series. *Mathematics of Computation*, 19(90):297–301, 1965. ISSN 1088-6842, 0025-5718. doi: 10.1090/S0025-5718-1965-0178586-1. URL <https://www.ams.org/mcom/1965-19-090/S0025-5718-1965-0178586-1/>.